

📍 India
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SUMMARY

Innovative Technology Professional with over 11 years of global experience. Recognized by 18 countries for contributions to technology, including 12 patents and 39 inventions. Delivered impactful results as a permanent member of JavaScript TC 39 and winner of several global awards including Innovator of the Year by CSIR, India in 2025 for outstanding invention. Dynamic Technical Lead Engineer skilled in Node.js, React.js, JavaScript, TypeScript, C++ and Python. Proven track record in enhancing user experiences and implementing innovative solutions. Committed to driving performance improvements through effective team leadership. I'm also the founder and developer of **Drishti.JS**. An open-source JS framework with inline CSS editor (supports SCSS), one-click patented UI feature to convert web app to Android/iOS app, inbuilt coding CLI with AI, both isolated and packaged Node.js.

SKILLS

- React.js and Next.js
- Node.js with Express
- JavaScript and TypeScript
- Redux management
- AWS and Azure. GCP (monitoring and management)
- Docker deployment
- MySQL, MongoDB
- PostgreSQL
- C++ and Python (NumPy, Pytorch and Lang chain)
- Agentive AI and Generative AI development
- Version control
- Jira project management

Dr Rahul Sarkar

EXPERIENCE

October 2023 - Current

Technical Lead Engineer and Architect at United Nations under Bhanja Minerals & Metals | Germany & India
Node with express and React with Next and migrations from PHP

- Migrated site from PHP to Node and React, enhancing performance and modernizing technology stacks.
- *Added custom modules using CORE JS*
- Developed payment gateways and APIs with decoupled security patches, improving security and transaction reliability.
- *Implemented AI chat bot. Designed the DBs.*
- Integrated in-house servers and implemented dual-stage hosting on AWS, increasing system resilience and scalability. Developed multiple REST APIs, FASTAPIs and GraphQL APIs.
- Developed subsets of AI into multiple GEN AI to tune servers and auto translate site content and quality. Report errors and fix them onsite.
- Used FIGMA to design the architectural flow.
- Git for CI/CD pipeline.

- *JavaScript, TypeScript, CSS (including Tailwind), HTML, MySQL and PostgreSQL, MongoDB, jQuery, Ajax, JIRA, Kanban, Azure,*
- *FIGMA, Power BI, Node.js with React, AWS*
- *Docker, Google Analytics and Tags, Kubernetes,*
- *CircleCi with dual channel GitHub.*
- *Quick JS*
- *C++ and Python (pytorch and Langchain)*

• <https://www.unhcr.org/>

• <https://unicef.org/>

• <https://www.citi.com/> (only payment section)

• <https://www.private-eye.co.uk/>

Open-source contributions

'Nov 2025

Promise.Pick ([Try it here!](#))

Others:

Created a Generative Artificial Intelligence for Drishti JS to handle, debug and analyze the code and framework architecture. Created AI Chat bot with multi user for same chat session.

Created Swap AI.

January 2021 - September 2023

Lead Engineer at IBM | Germany

Node with Express and Next. CORE JS

- Led team while contributing individually to development of key features.
- Developed additional payment features to enhance user transaction experience.

- Power BI analytics
- PHP
- Golang, REST, FASTAPIs
- Office productivity
- Figma and Adobe design tools

- Integrated QuickJS as an additional engine for custom JavaScript functionality.
- Additional sub user handling with regional based.
- Implemented GraphQL and Rest async way.
- Customization on AWS. Integrated developed projects Next JS domain APIs and Clubbed APIs (partial with REST).
- Added custom performance logging mechanism.
- Implemented architectural microservices.
- Developed multiple REST APIs, FASTAPIs and GraphQL APIs.
- Built an inhouse AI to monitor and report server errors, logs, code bottlenecks and added additional features for GEN AI.
- Developed an AI for a healthcare application
 - Used FIGMA and bit of Adobe to design the architectural flow.
 - Git for CI/CD pipeline.

Technologies used:

- JavaScript, Typescript, CSS + SCSS, HTML.
 - MongoDB/MySQL, jQuery, Ajax, Acquia, Node JS coupled with Next, Typescript, AWS, Google Cloud, Google Analytics, Docker and Tags.
- For AI/GEN AI: C++ WITH PYTHON (pytorch and Langchain). FASTAPI
 Provided links to significant project websites for reference.

- <https://www.tourradar.com/>
- <https://www.washingtonpost.com/>
- <https://www.bloomberg.com/>
- <https://www.walmart.com/> (for Italy and Portugal only).

Open-source contributions:

- Part of Loopback, an open-source framework to build dynamic ETE REST Api.
- Open API to GraphQL, helps translating APIs from Open API to GraphQL.

December 2014 - December 2021

Deputy Manager at Coal India Limited (worked in subsidiary Central Mine Planning and Design Institute Ltd) | India & Ethiopia

- Designed and developed applications for the ministry of coal, government of India, enhancing service delivery.
- Contributed to development of applications as part of the NIC (National Informatics Centre) team, supporting various governmental initiatives.
- Led team of junior members, mentoring and guiding them in application development projects.
- Developed low scale applications using C++.

Utilized various technologies for application development.

- *Php*
- *Data integration using Python*
- *Minimal Gen AI* (for Mining equipment monitoring)

Project URL:

1. <https://www.cmpdi.co.in/en>
2. <https://www.coalindia.in/>

EDUCATION AND TRAINING

September 2019

Ph.D. | Advance Computers and Electronics
Technical University of Munich, Germany
CS (D), Silver Medalist

June 2014

Bachelor of Engineering Technology | Electrical and Electronics
Engineering
Andhra University, India

LANGUAGES

English:	Proficient	Bengali:	Native
German:	Intermediate	Hindi:	Proficient
Telugu:	Elementary		

RECENT AWARDS & HONORS

CSIR, 2025- Innovator of the Year, Govt of India
Red Hat, 2024- Best contributor

BRICS, 2021- Awarded for enhancing Ethiopia's rural technological sector

CERTIFICATIONS & LICENSES

AWS

Several patents

ACTIVITIES

Part of National rural skilled provider, India